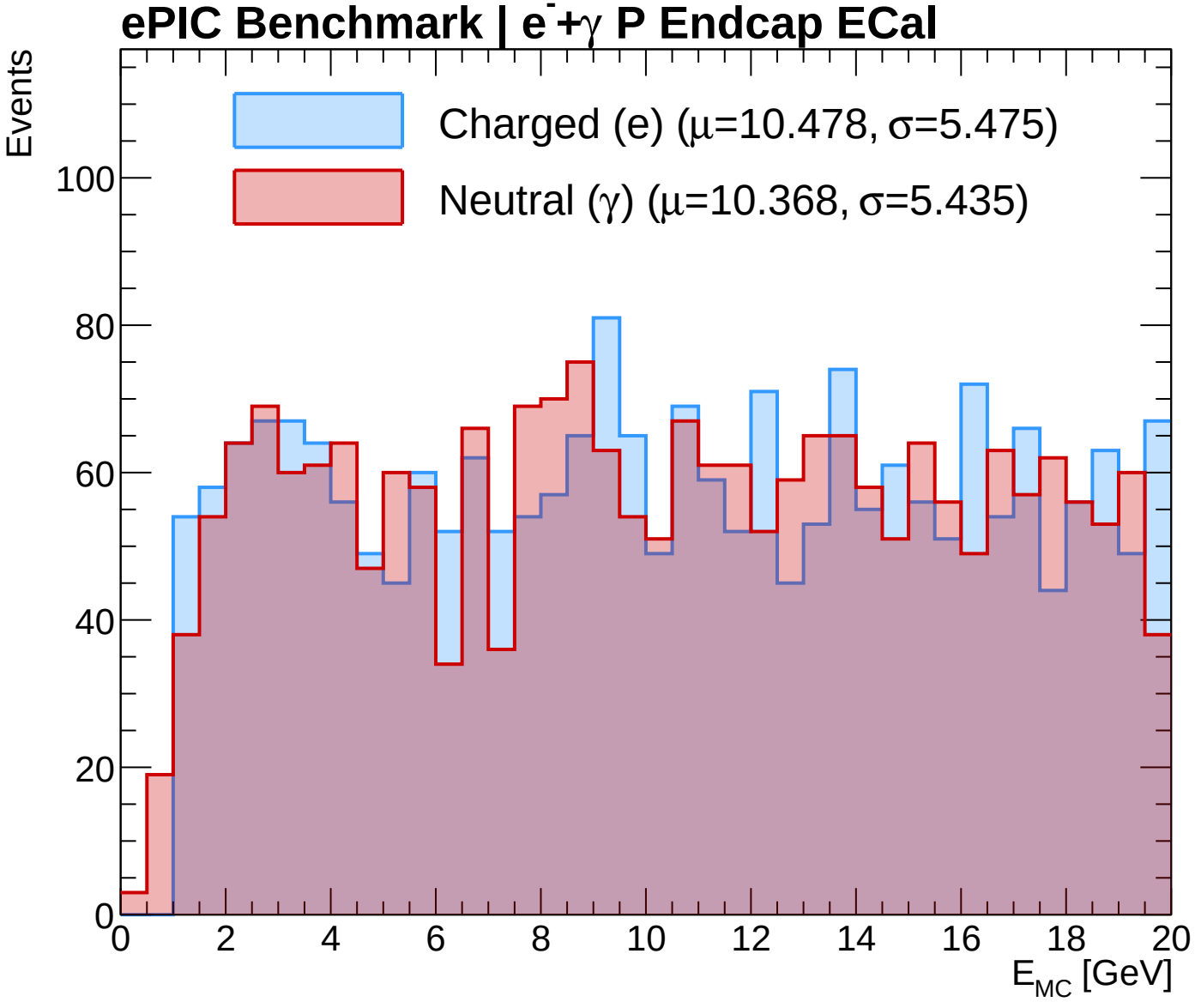
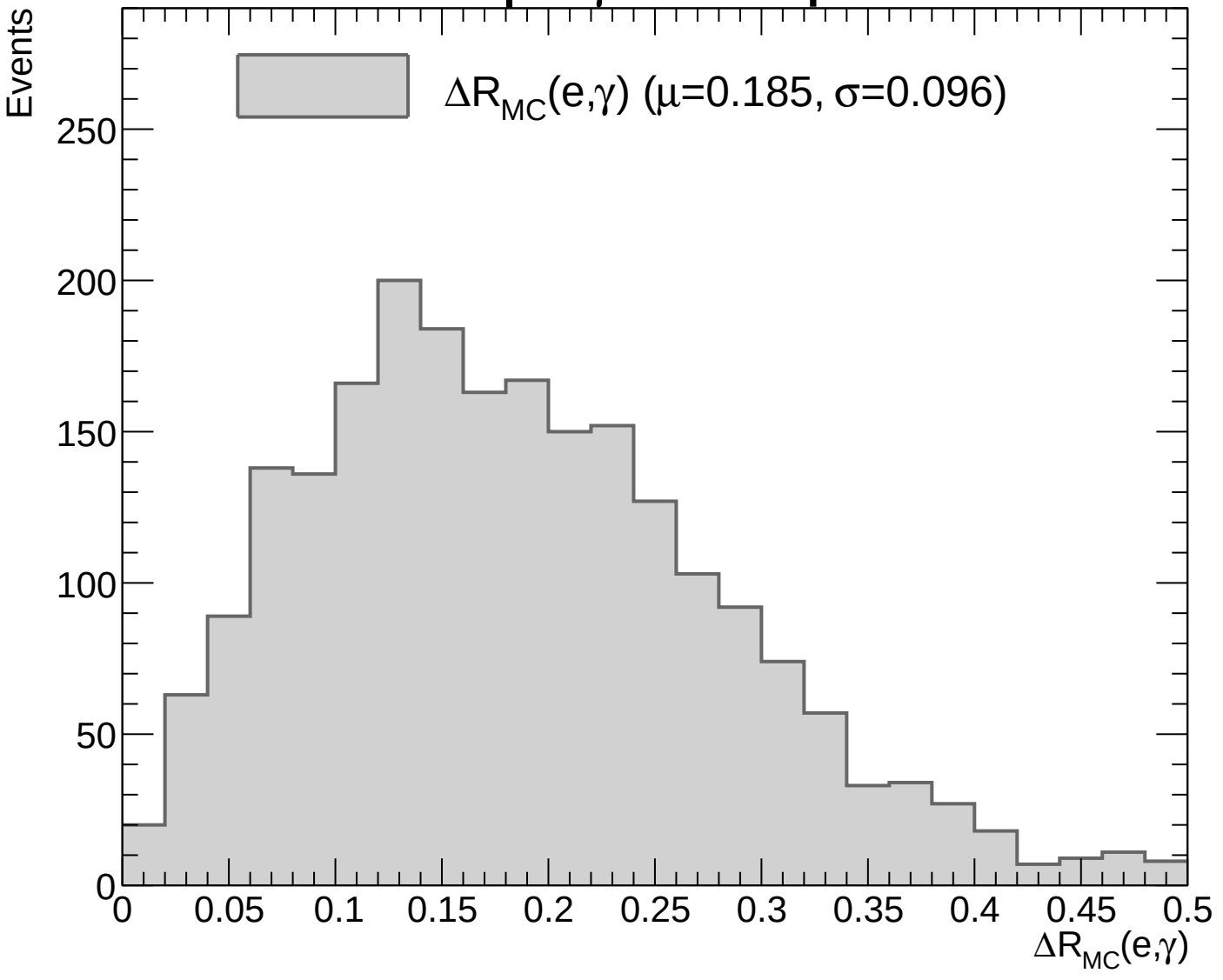
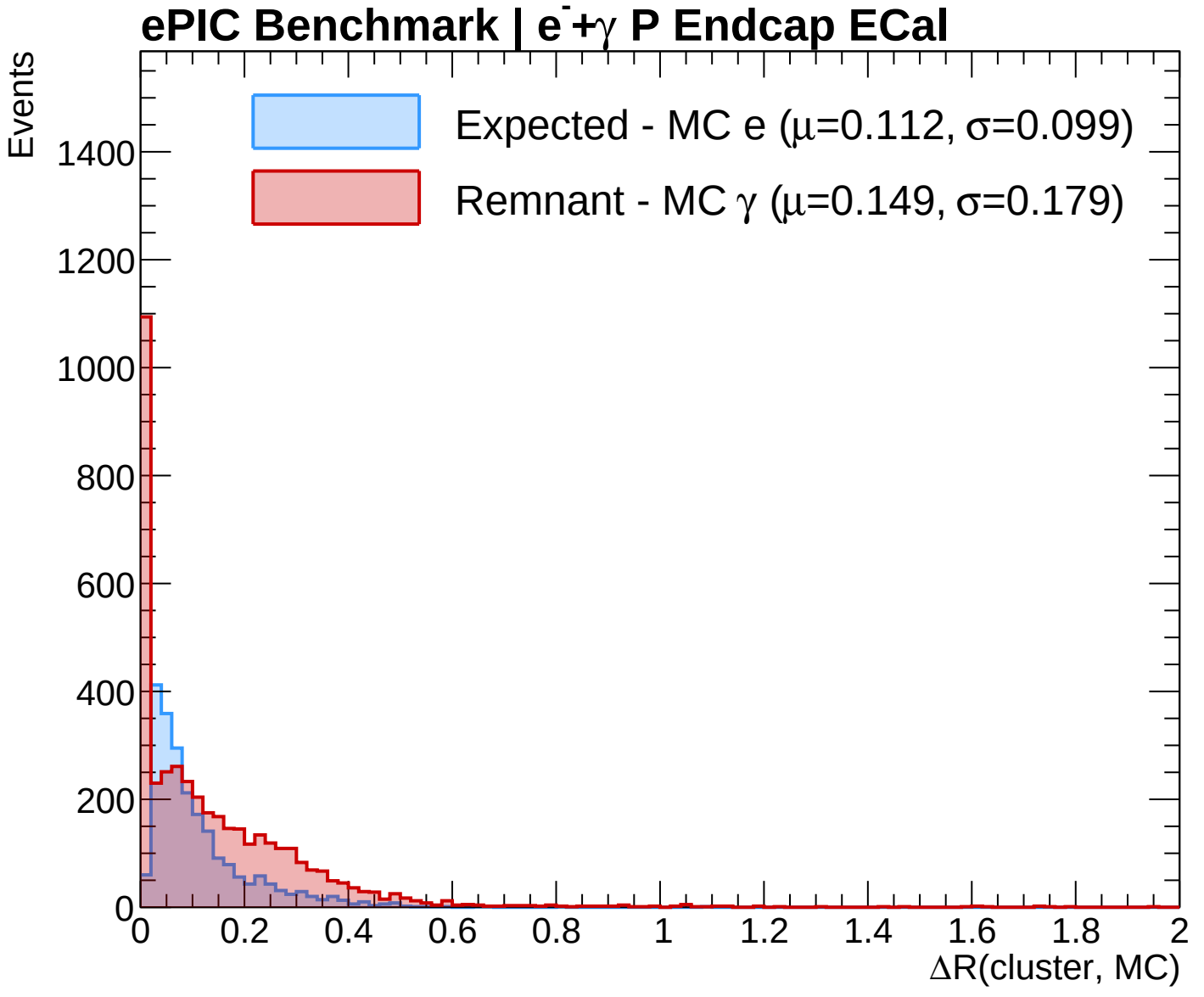
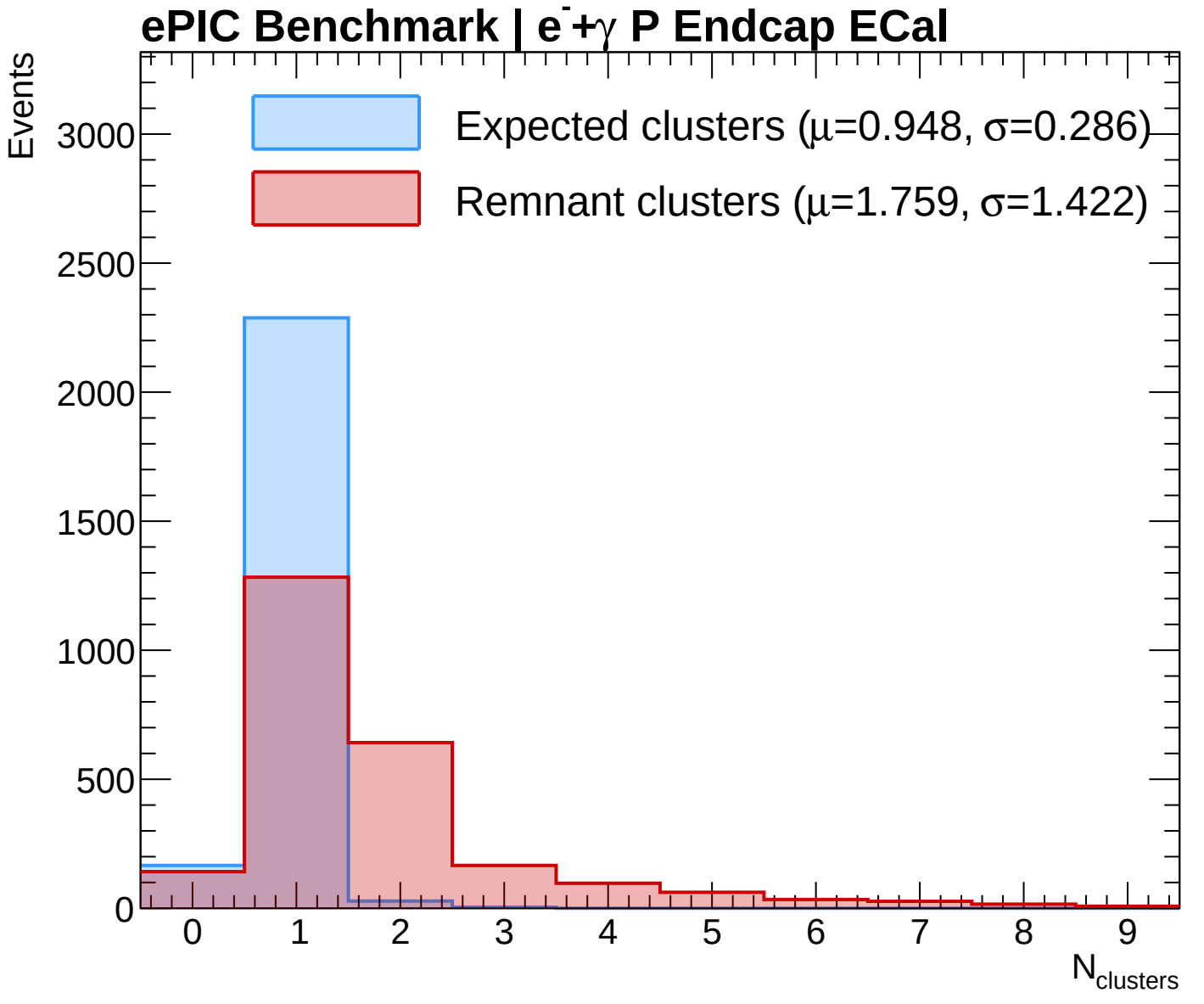
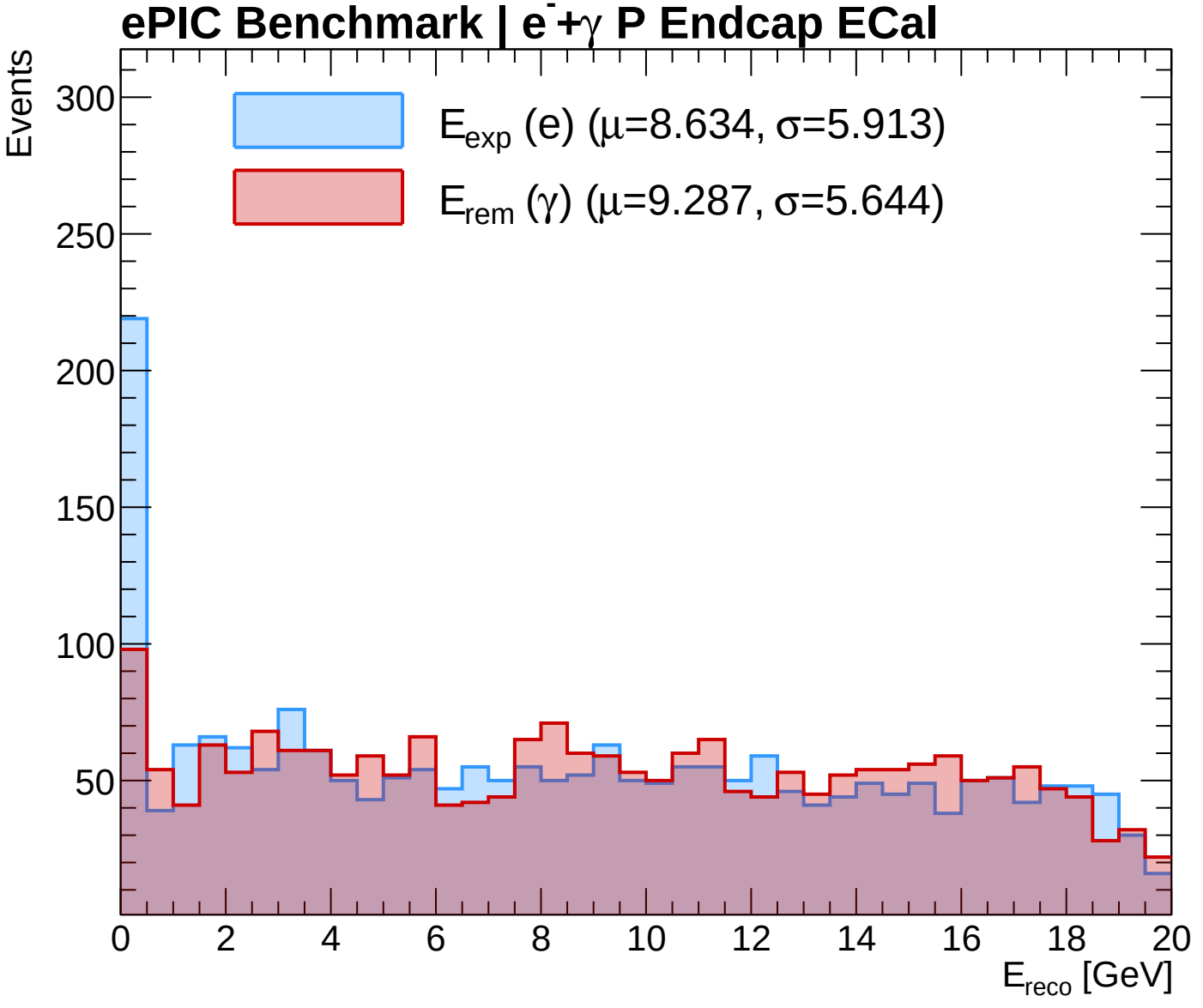
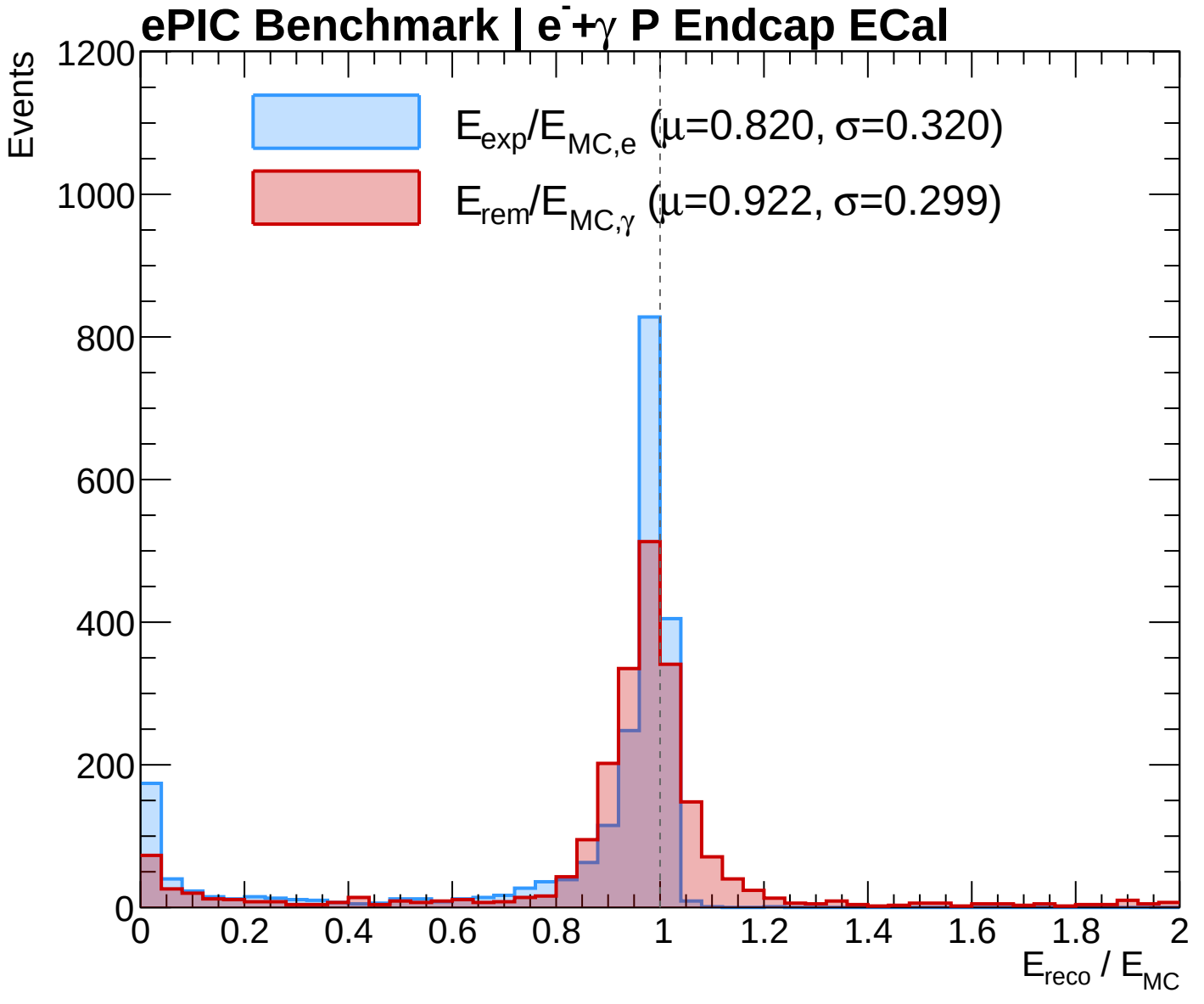

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
Charged (e) (μ=10.478, σ=5.475)
Neutral (γ) (μ=10.368, σ=5.435)
0
20
40
60
80
100
0
2
4
6
8
10
12
14
16
18
20
E_MC [GeV]

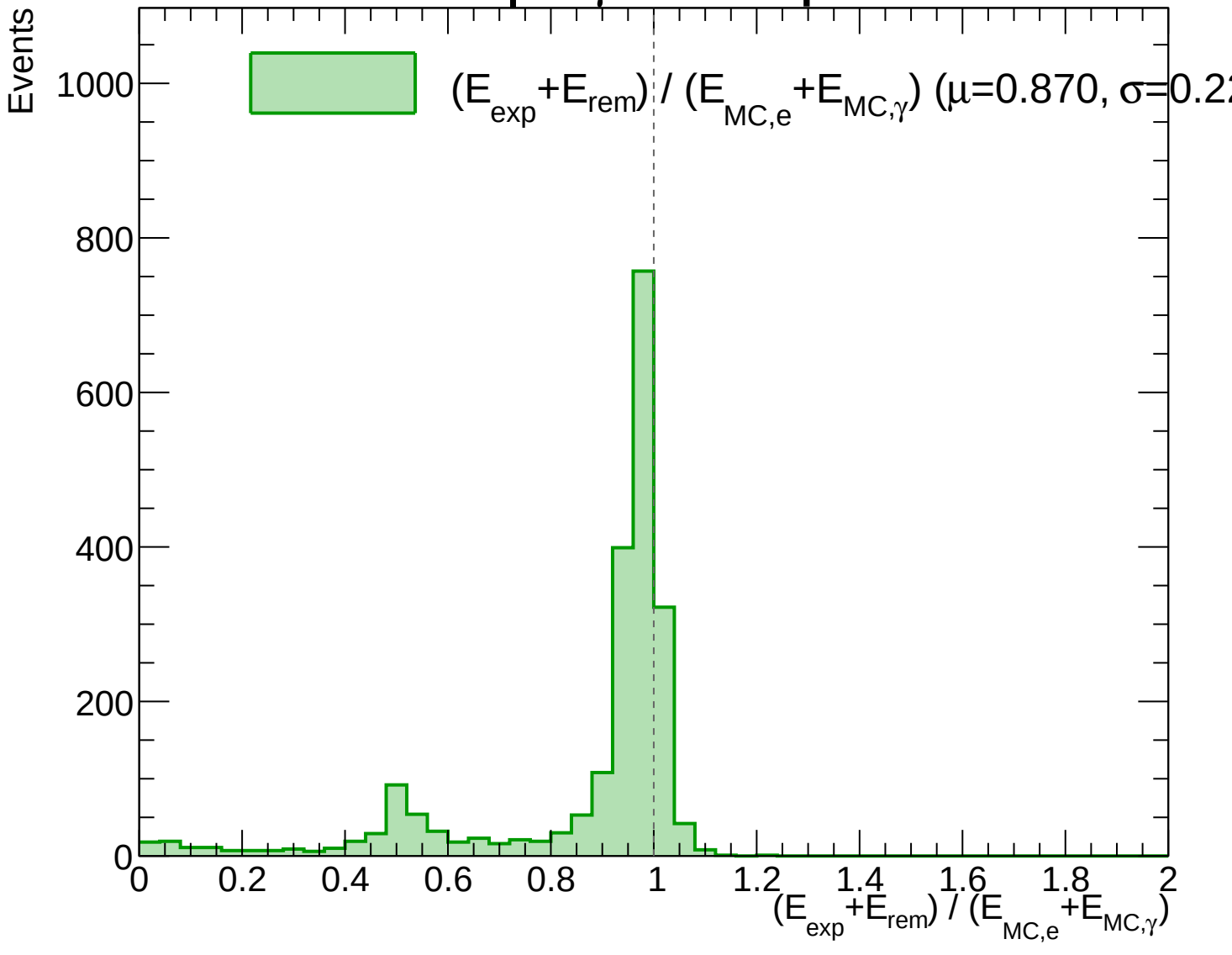

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
ΔR_MC(e,γ) (μ=0.185, σ=0.096)
250
200
150
100
50
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)

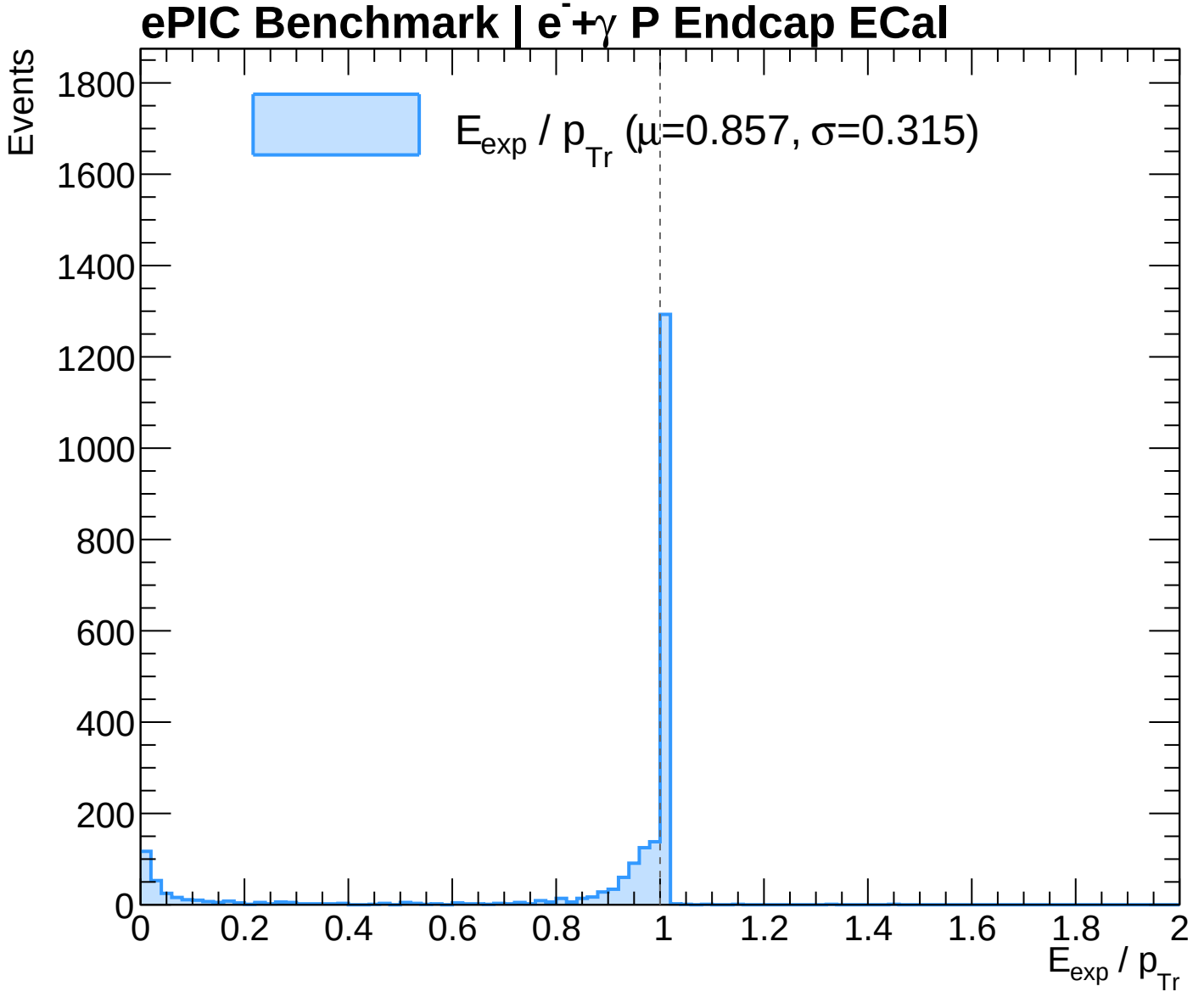

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
Expected - MC e (μ=0.112, σ=0.099)
Remnant - MC γ (μ=0.149, σ=0.179)
0
200
400
600
800
1000
1200
1400
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
ΔR(cluster, MC)

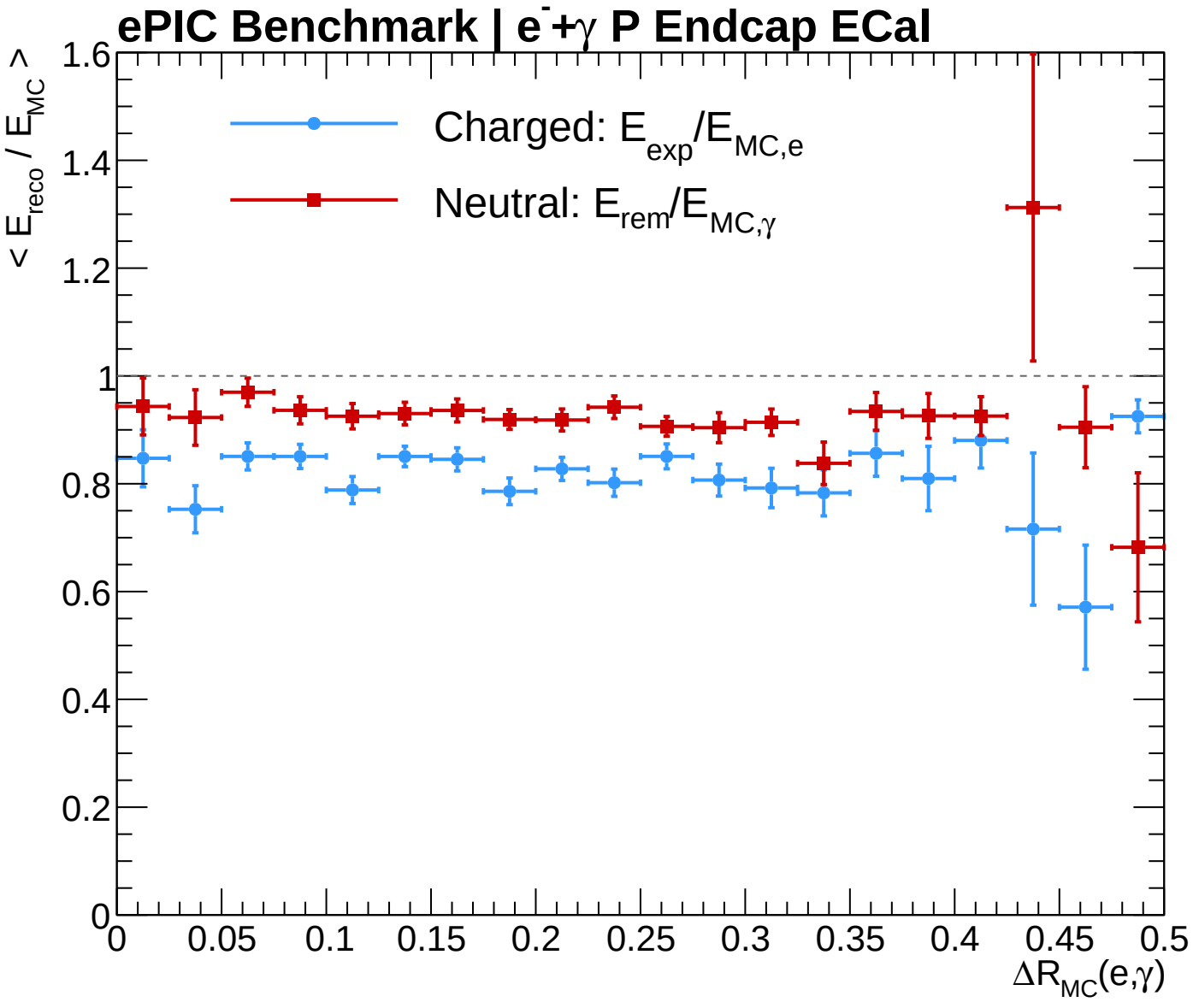

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
3000
2500
2000
1500
1000
500
0
0
1
2
3
4
5
6
7
8
9
N clusters
Expected clusters (μ=0.948, σ=0.286)
Remnant clusters (μ=1.759, σ=1.422)

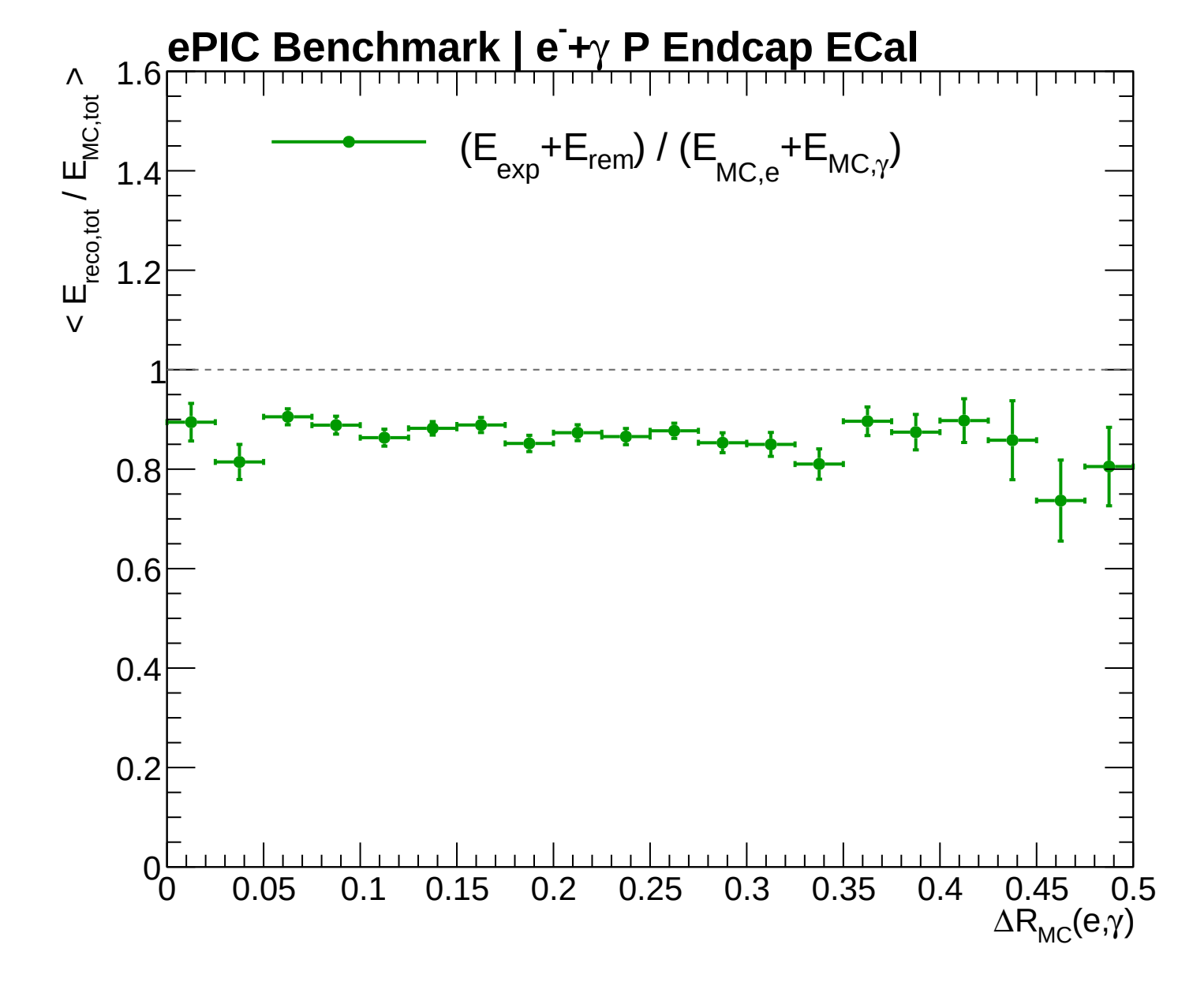

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
300
250
200
150
100
50
0
2
4
6
8
10
12
14
16
18
20
E_reco [GeV]
E_exp (e) (μ=8.634, σ=5.913)
E_rem (γ) (μ=9.287, σ=5.644)

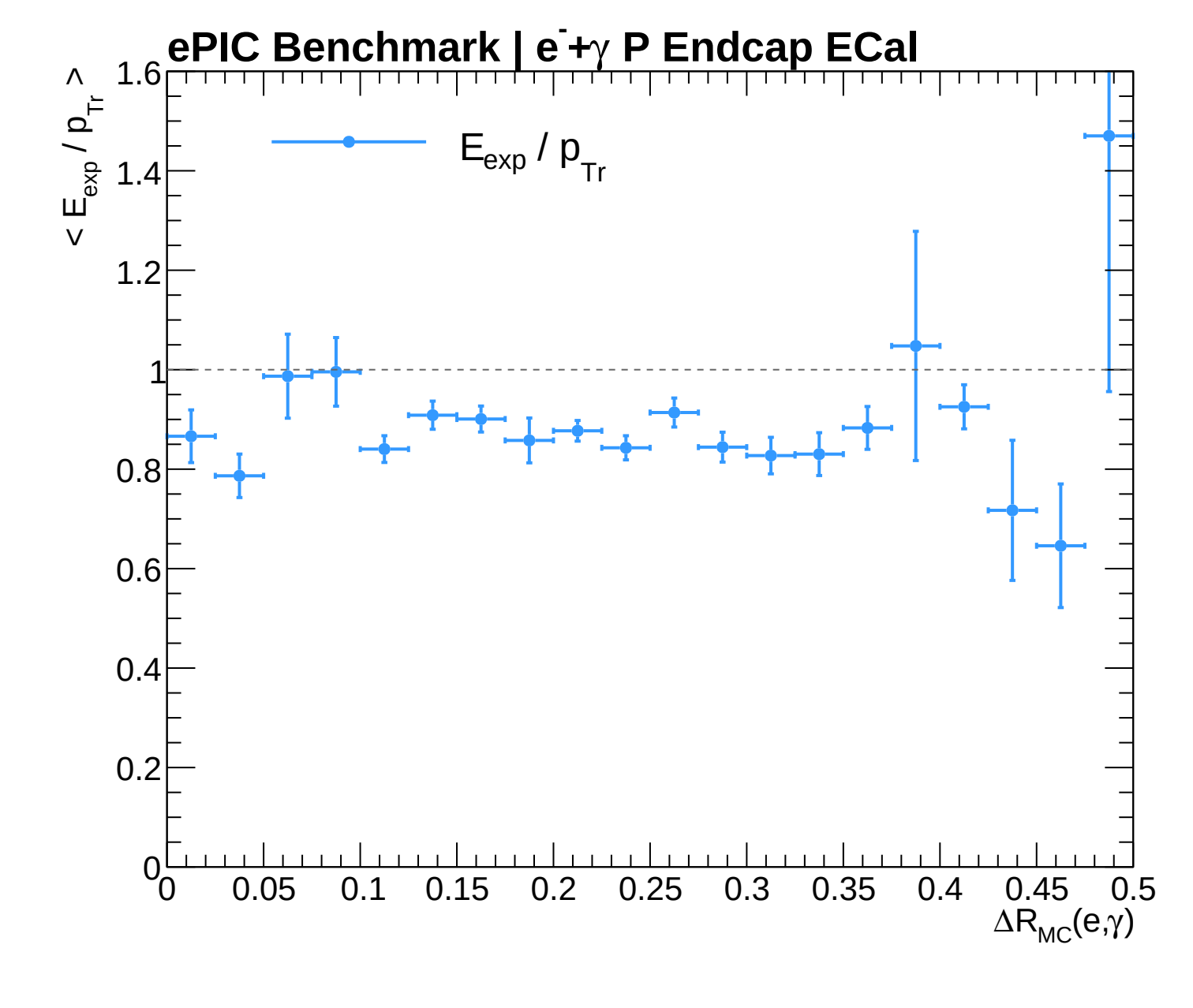

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
1200
1000
800
600
400
200
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_exp/E_MC,e (μ=0.820, σ=0.320)
E_rem/E_MC,γ (μ=0.922, σ=0.299)
E_reco / E_MC

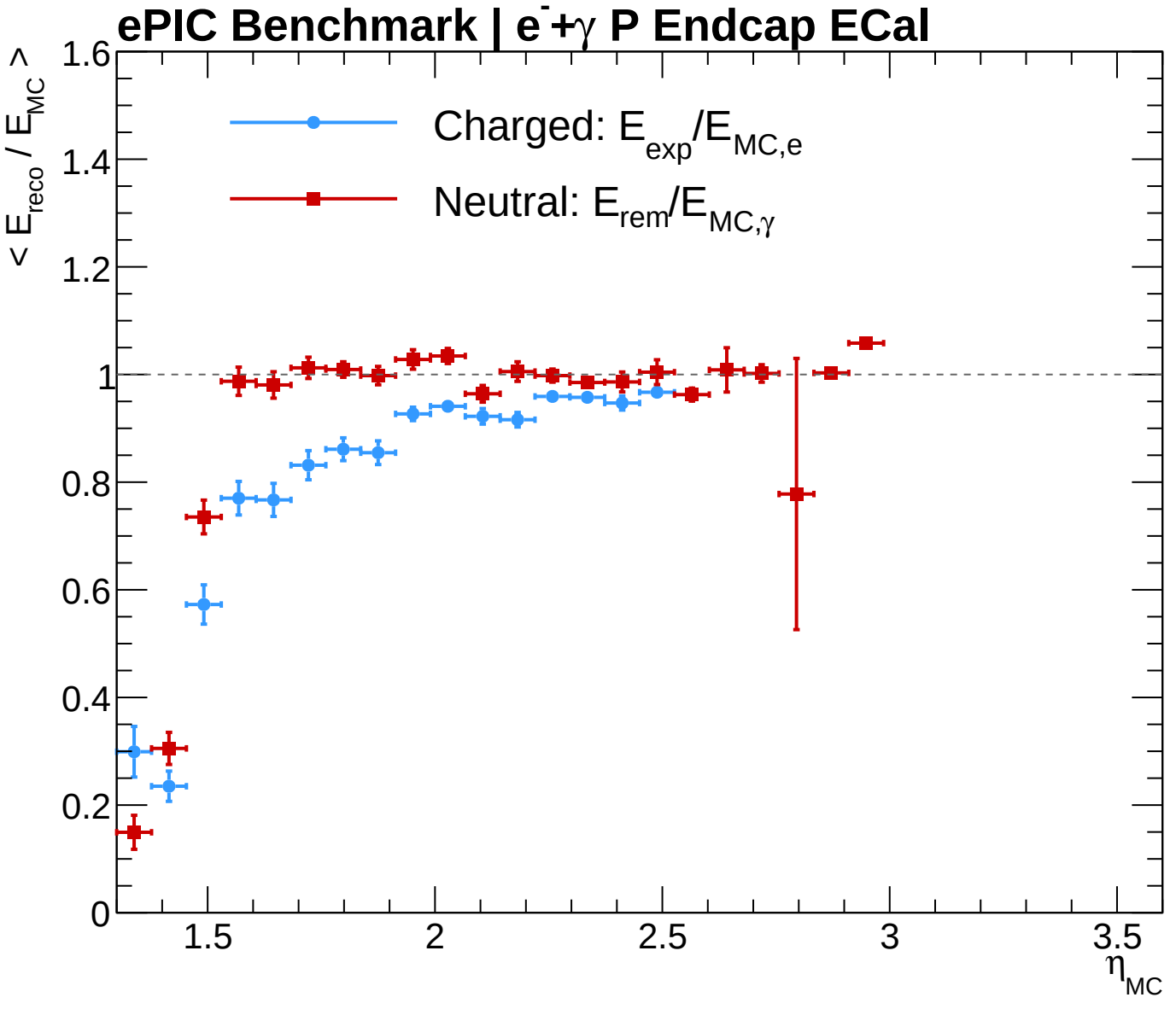

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
(E_exp+E_rem) / (E_MC,e+E_MC,γ) (μ=0.870, σ=0.22
1000
800
600
400
200
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
(E_exp+E_rem) / (E_MC,e+E_MC,γ)

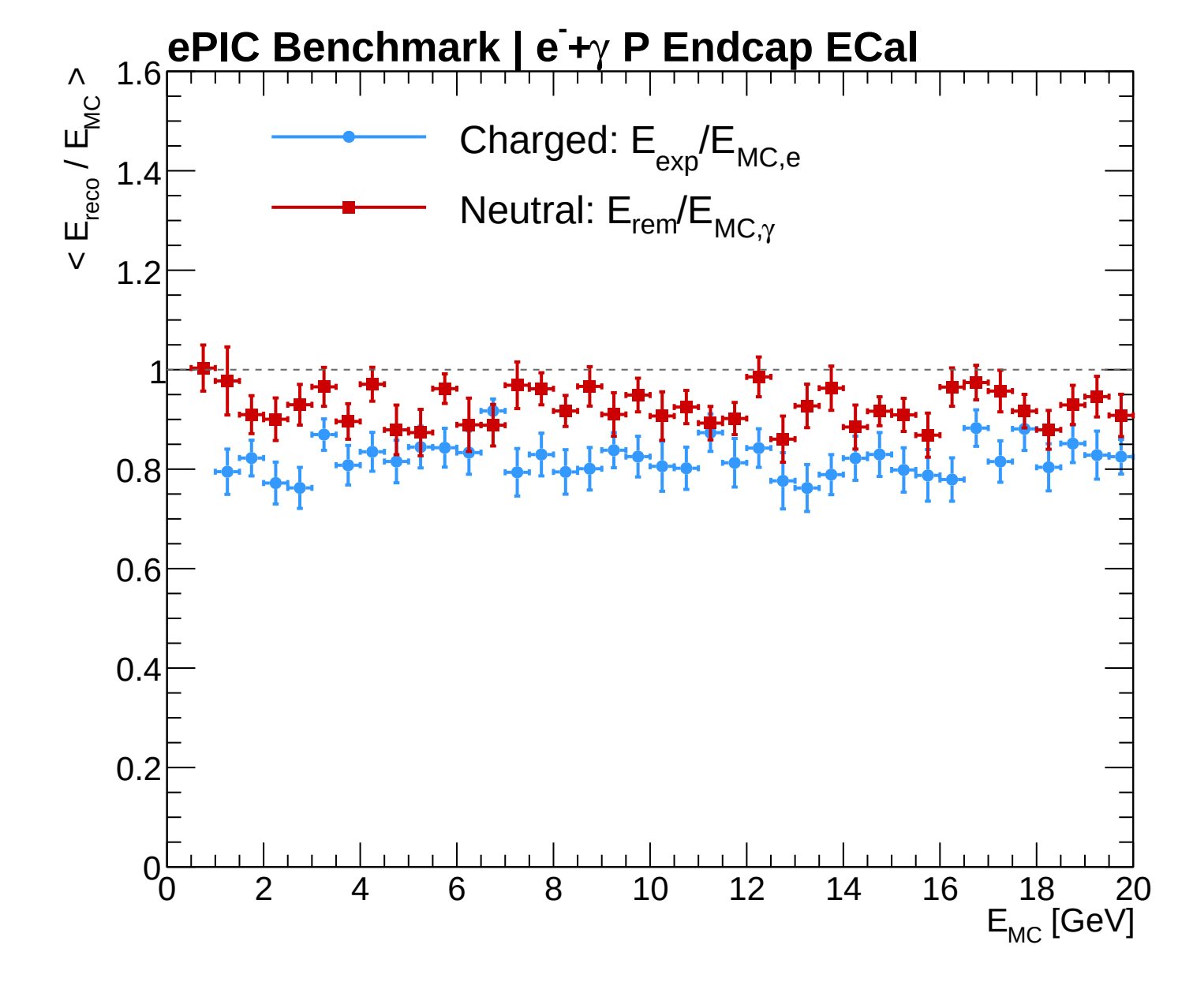

ePIC Benchmark | e⁻+γ P Endcap ECal
Events
E_exp / p_Tr (μ=0.857, σ=0.315)
1800
1600
1400
1200
1000
800
600
400
200
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_exp / p_Tr

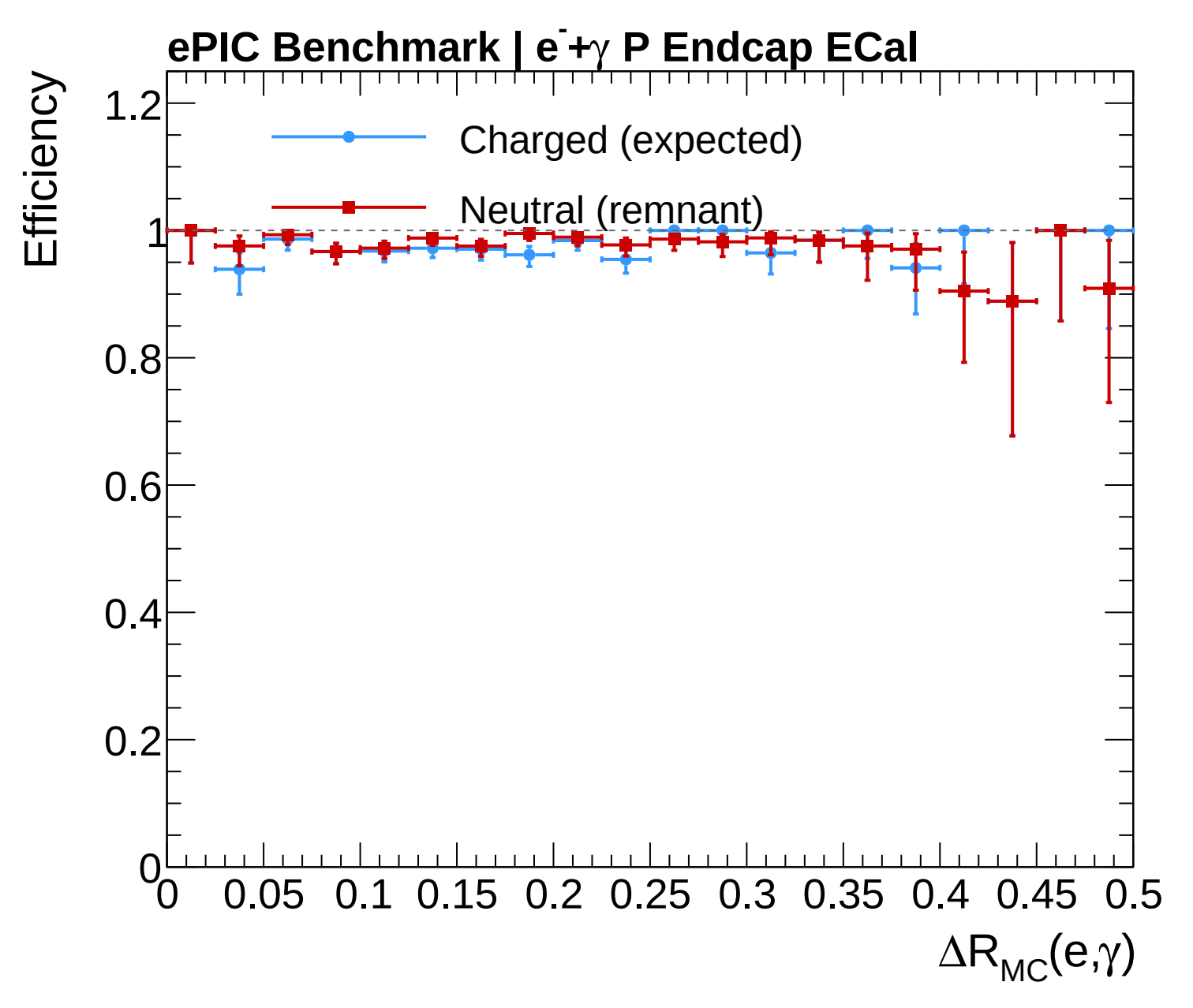

ePIC Benchmark | e⁻+γ P Endcap ECal
< E_reco / E_MC >
Charged: E_exp/E_MC,e
Neutral: E_rem/E_MC,γ
ΔR_MC(e,γ)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5

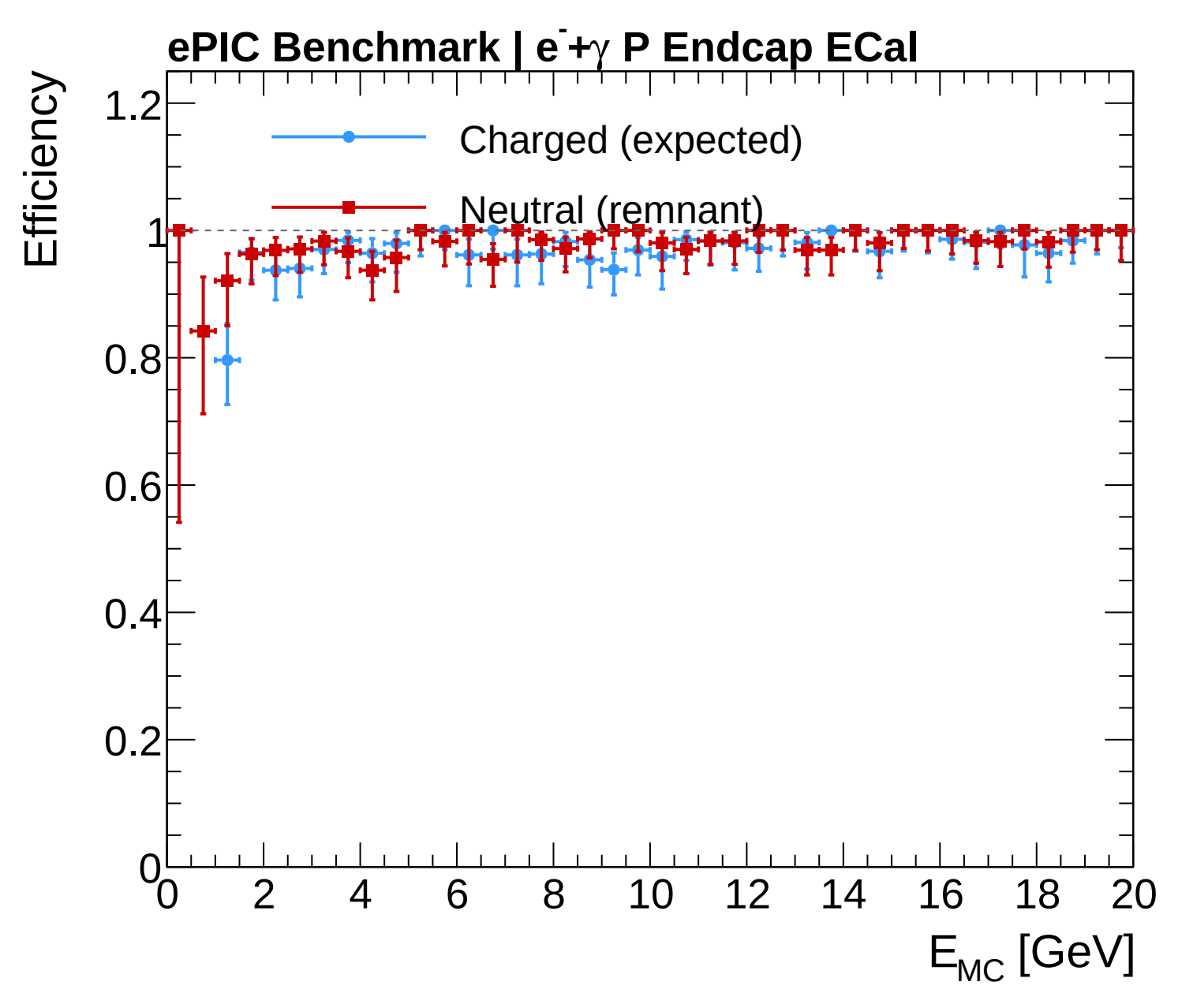

ePIC Benchmark | e⁻+γ P Endcap ECal
(E_exp+E_rem) / (E_MC,e+E_MC,γ)
< E_reco,tot / E_MC,tot >
ΔR_MC(e,γ)
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5

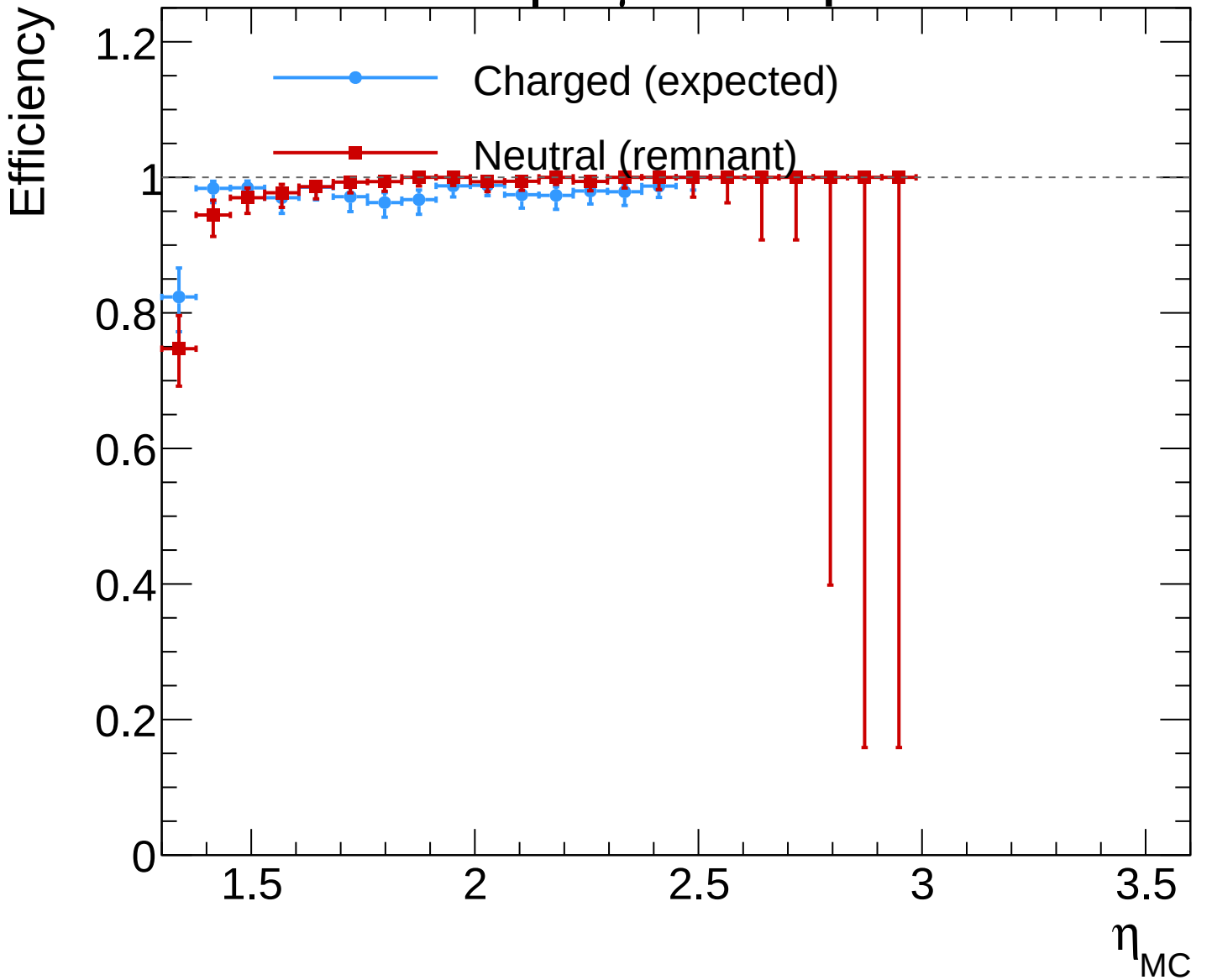

ePIC Benchmark | e-+γ P Endcap ECal
< E_exp / p_Tr >
E_exp / p_Tr
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)


ePIC Benchmark | e⁻+γ P Endcap ECal
< E_reco / E_MC >
Charged: E_exp/E_MC,e
Neutral: E_rem/E_MC,γ
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
1.5
2
2.5
3
3.5
η_MC


ePIC Benchmark | e-+γ P Endcap ECal
< Ereco / EMC >
Charged: Eexp/EMC,e
Neutral: Erem/EMC,γ
EMC [GeV]
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
0
2
4
6
8
10
12
14
16
18
20


ePIC Benchmark | e⁻+γ P Endcap ECal
Efficiency
Charged (expected)
Neutral (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
ΔR_MC(e,γ)


ePIC Benchmark | e⁻+γ P Endcap ECal
Efficiency
Charged (expected)
Neutral (remnant)
1.2
1
0.8
0.6
0.4
0.2
0
0
2
4
6
8
10
12
14
16
18
20
E_MC [GeV]


ePIC Benchmark | e⁻+γ P Endcap ECal
Efficiency
1.2
1
0.8
0.6
0.4
0.2
0
Charged (expected)
Neutral (remnant)
1.5
2
2.5
3
3.5
η_MC